

ACTIVITY NO. 5

DIFFRACTION PATTERN USING A LASER BEAM

Aim: To determine the grating element (d) of the given grating.

Apparatus: Diffraction grating, laser pointer, stands, screen and meter Scale.

Formula:

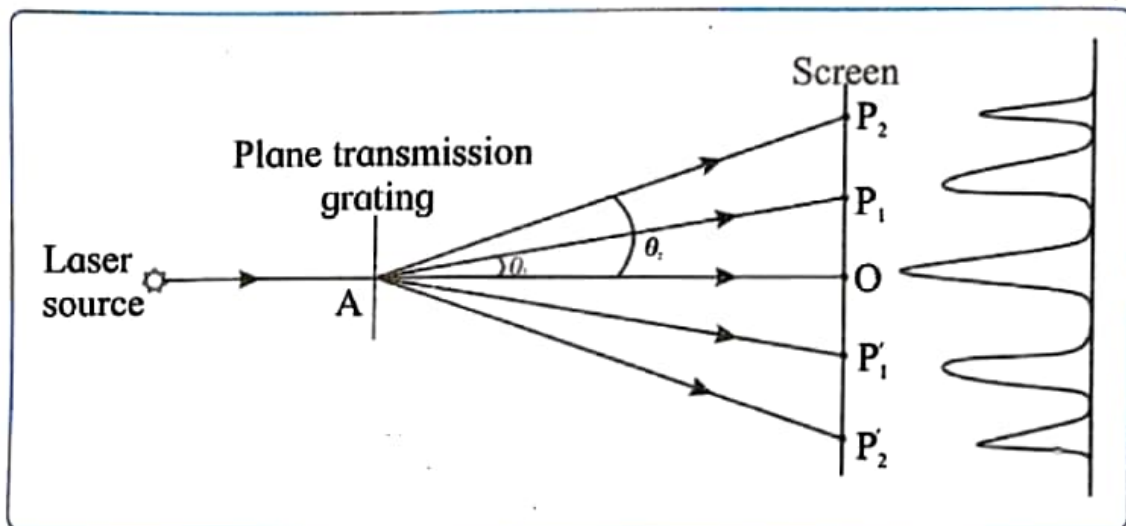
$$n\lambda = d \sin \theta,$$

Where, λ = the wavelength of the laser light,

d = grating element

and θ = the angle of diffraction.

Diagram :



Diffraction by plane transmission grating

Intensity distribution curve

Theory :

Diffraction grating:- suppose that instead of a single slit, or two slits side by side as in young's experiment, we have a very large number of parallel slits all of the same width and spaced at regular intervals, the arrangement is known as diffraction grating. It was first constructed by Fraunhofer. Gratings are made by ruling, with a diamond point, a large number of equidistant grooves on a glass or metal surface. The distance between consecutive slits of a grating is its grating element (d).

Procedure :

1. Adjust the laser pointer (red or green) on its stand and switch it on.
2. Mount the grating on the retort stand and place it in the path of the laser beam. The position of the grating is so adjusted, that the plane of the grating is nearly perpendicular to the laser beam.
3. Red spots appear on the screen. These spots represent the diffraction maxima of various orders. If the spots are not symmetrical about the central spot, adjust the plane of the grating to make them so.

4. Measure the distance D between the grating and the screen. (Distance AO in the diagram).
5. Measure the distance L of the central spot from the first and second spot on either side of it. (Distances OP_1 , OP_2 , OP_1' , OP_2' in the diagram.)

Observations :

1. Wavelength of the laser light used (red) : $\lambda = 6328 \text{ \AA} = 6.328 \times 10^{-5} \text{ cm}$

Observation Table :

Set no	Distance between grating and screen D (cm)	Order number	Distance between the central spot and nth spot			$\tan \theta = (L/D)$	$\theta = \tan^{-1} (L/D)$	$d = \frac{n\lambda}{\sin \theta}$ (cm)
			L_1	L_1'	mean			
1	3	1	1.3	1.4	1.35	0.45	24.2°	15.43×10^{-5}
	6	2	2.6	2.7	2.65	0.44	23.5°	15.68×10^{-5}

Note the Experiment can be done by single slit method using $D > 1.5 \text{ m}$

Result :

The grating element of the diffraction grating is $15.555 \times 10^{-5} \text{ cm}$

Precautions:

1. Do not look directly into the laser beam.
2. Adjust the grating such that the diffraction pattern is symmetric about the central maxima.

Questions

1. What do you mean by diffraction of light?

Diffraction is the phenomenon of bending of light around corners of an obstacle in the path of light.

2. Can we get a diffraction grating in our daily life?

- 1) Holograms
- 2) X-ray diffraction
- 3) Sun appears red during sunset
- 4) CD reflecting rainbow colours.
- 5) To separate white light.

Remark and sign of teacher :